



GROWTH & GH AXIS

PEG-MGF

PEG-MGF – 2 mg per vial – 10 vials per box

FOR RESEARCH USE ONLY (RUO). NOT FOR HUMAN OR VETERINARY USE.

IDENTIFICATION

CHEMICAL NAME	PEGylated mechano growth factor
SYNONYMS / ALIASES	PEG-MGF; pegylated IGF-1Ec C-terminal peptide
CAS NUMBER	[VERIFY WITH COA]
MOLECULAR FORMULA	MGF peptide core + polyethylene glycol moiety (variable PEG mass) [VERIFY WITH COA]
MOLECULAR WEIGHT	MGF core ~2867 g/mol plus PEG (total varies by PEG size) [VERIFY WITH COA]
AMINO ACID SEQUENCE	YQPPSTNKNTKSQRRKGSTFEERK (IGF-1Ec C-terminal E-domain) with covalent N-terminal polyethylene glycol conjugation
SOURCE	Synthetic (solid-phase peptide synthesis with PEG conjugation)

PHYSICAL PROPERTIES

APPEARANCE	Lyophilized white to off-white powder
PURITY SPECIFICATION	>=98% by HPLC (confirm against batch COA)
CONTAINER	2 mg per vial, sealed glass vial, rubber-stoppered, aluminum crimp

STORAGE & HANDLING

UNRECONSTITUTED STORAGE	-20 C, protect from light and moisture
RECONSTITUTED STORAGE	2-8 C, stable approximately 14-21 days in bacteriostatic water (PEGylation extends solution stability vs. native MGF)
SUGGESTED RECONSTITUTION SOLVENT	Bacteriostatic water or sterile water (research use only)
SHELF LIFE (UNRECONSTITUTED, SEALED)	24 months at -20 C
STABILITY NOTES	Avoid repeated freeze-thaw cycles. PEG conjugation increases hydrodynamic radius and resistance to proteolysis relative to unmodified [[mgf]].

STRUCTURAL NOTES

PEG-MGF is the polyethylene-glycol-conjugated form of mechano growth factor, the C-terminal E-domain peptide of the IGF-1Ec splice variant. Covalent attachment of a PEG moiety increases molecular size and shields the peptide from rapid proteolytic degradation, substantially extending its half-life in solution relative to the unmodified [[mgf]] peptide. The peptide core sequence is unchanged by PEGylation.

SELECTED PEER-REVIEWED REFERENCES

1. Yang S., Goldspink G. - *Different roles of the IGF-1Ec peptide (MGF) and mature IGF-1 in myoblast proliferation and differentiation* - FEBS Letters, 2002 - DOI: 10.1016/S0014-5793(02)03742-7
2. Veronese F.M., Pasut G. - *PEGylation, successful approach to drug delivery* - Drug Discovery Today, 2005 - DOI: 10.1016/S1359-6446(05)03575-0

References are provided for research background only. Cited publications do not constitute claims about the safety, efficacy, or intended use of this product.

COMPLIANCE DISCLAIMER

This product is sold by Aligned Aminos for laboratory research use only (RUO). It is not a drug, dietary supplement, food, cosmetic, or medical device. It has not been approved by the FDA for any therapeutic, diagnostic, or preventative use. It is not intended for use in humans or animals, and may not be administered, ingested, applied, or otherwise introduced into any living organism. Purchasers represent that they are qualified researchers or laboratory professionals and assume full responsibility for compliance with all applicable laws and institutional review requirements governing the handling of research chemicals.

Batch-specific identity and purity are documented in the Certificate of Analysis (COA) issued for the lot.